KENTARO VADNEY

Python Developer | ML Pipelines & Distributed Computing

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Python Developer with 3+ years of production experience building scalable ML pipelines, distributed systems, and high-performance data processing infrastructure. Expert in Python optimization, distributed computing, and cloud-native ML workflows. Reduced database latency 60% and supported 5x throughput increases through algorithmic improvements and system design. Building ML systems at Mercor, seeking to architect large-scale ML pipelines.

TECHNICAL SKILLS

Languages: Python (best), JavaScript/TypeScript, SQL, Java, Rust, Solidity

Distributed Computing: Kubernetes, Docker, Microservices, Message Queues, Consensus Protocols

Databases: MongoDB, PostgreSQL, MySQL, Redis, Query Optimization, Schema Design, Performance Tuning

ML & Data Processing: PyTorch, Dask, Pandas, NumPy, Scikit-learn, ML Pipeline Design, Model Deployment

Cloud Platforms: AWS (EC2, S3, Lambda, Kendra), GCP, Azure, Terraform, CI/CD (GitHub Actions, CircleCI)

Web Frameworks: FastAPI, Flask, Django, React, Node.js, Express, RESTful APIs, GraphQL

EXPERIENCE

Machine Learning Engineer • Mercor — Dec 2025 – Present • Remote (Contract)

- Engineered high-quality ML training datasets for LLM fine-tuning by annotating agentic trajectories, implementing solutions using PyTorch, LightGBM, XGBoost, and scikit-learn with 5-fold cross-validation on large-scale Kaggle datasets
- Validated SWE-Bench Pro dataset for autonomous code agents, analyzing model failure patches against golden solutions to audit test fairness, identify legitimate failures, and improve LLM evaluation benchmarks
- Developed golden responses and evaluation rubrics for AI agent training data, designing multi-step reasoning tasks with structured prompts, data synthesis pipelines, and domain-specific feature engineering workflows

Full Stack Software Engineer • Centigrade — Apr 2025 – Aug 2025 • Santa Clara, CA

- Architected high-performance API services with FastAPI and Python, implementing advanced query optimization and caching strategies that supported 5x throughput increase and reduced database latency 60%
- Built AWS Kendra AI search system with Python-based indexing pipeline processing 10,000+ documents, reducing manual discovery time 10+ hours/week through intelligent document retrieval
- Optimized data processing workflows using Pandas and NumPy, improving ETL pipeline performance 3x through vectorization and parallel processing

Founding Software Engineer • CarbonSustain — May 2024 – Dec 2024 • Remote

- Designed distributed transaction processing system with Python and Node.js, implementing real-time validation pipelines for blockchain-based carbon credit tracking across 3 enterprise deployments
- Built scalable microservices architecture with FastAPI, implementing RESTful APIs and asynchronous task queues handling 1000+ transactions/day

R&D Software Engineer • Belden — Jul 2023 – May 2024 • San Jose, CA

- Developed ML-powered threat detection pipelines using Python, integrating LLM models for industrial IoT cybersecurity analysis with Kubernetes deployment at scale
- Optimized machine learning model inference pipeline, reducing detection latency 40% through algorithm refinement
- Enhanced threat detection models expanding coverage 30% through feature engineering, data augmentation, and model architecture improvements
- Built containerized Python services with Docker and Kubernetes, implementing CI/CD pipelines for automated model deployment and version control

Additional Experience: Research & Blockchain (2021-2023)

- **UC Berkeley RISE Lab:** Led distributed systems research; implemented consensus algorithms in Python/Rust; co-authored peer-reviewed paper on Byzantine fault tolerance

EDUCATION

B.A., Computer Science • University of California, Berkeley — Aug 2019 – May 2023

- Regents' and Chancellor's Scholar (top 1% merit scholarship)